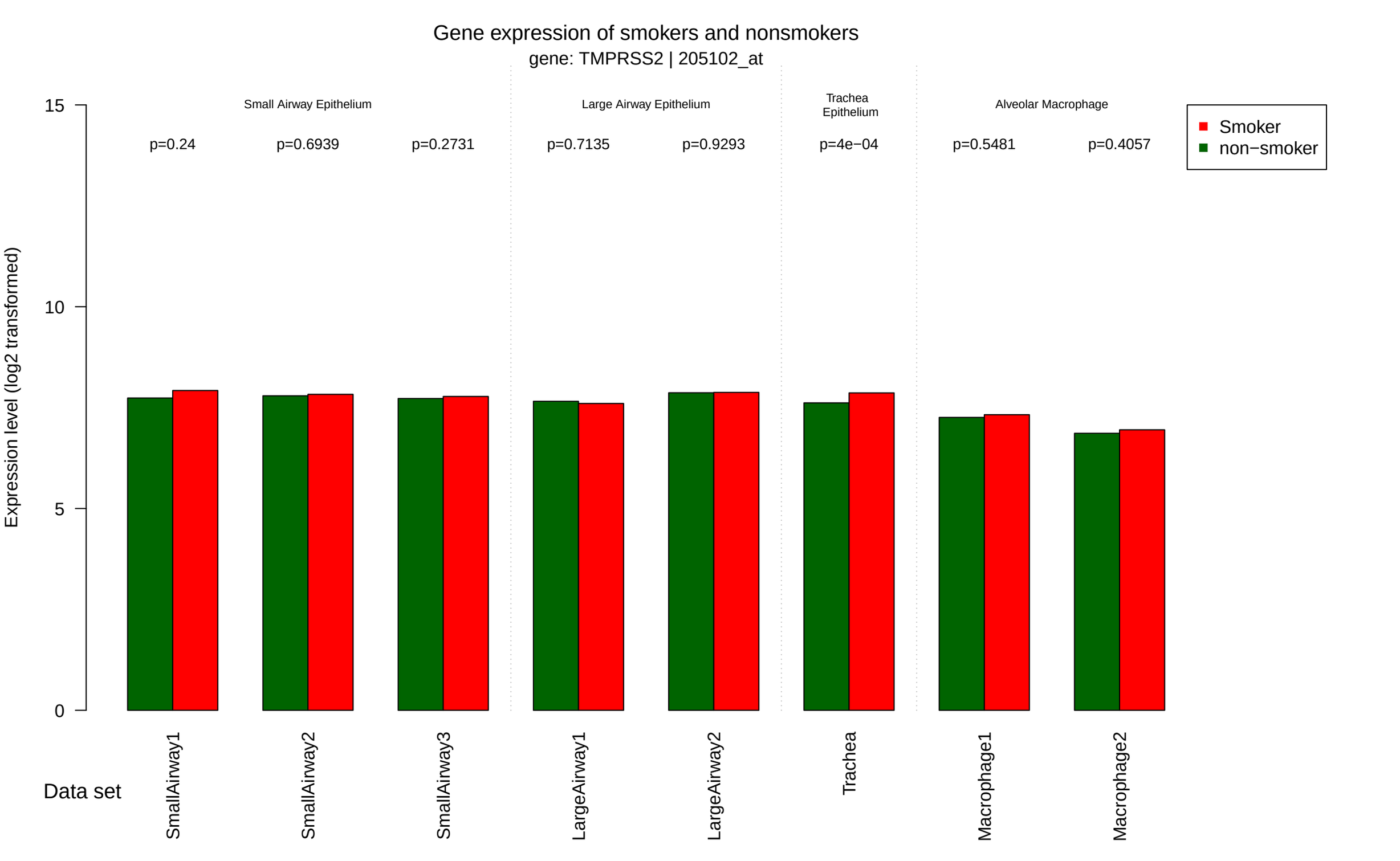
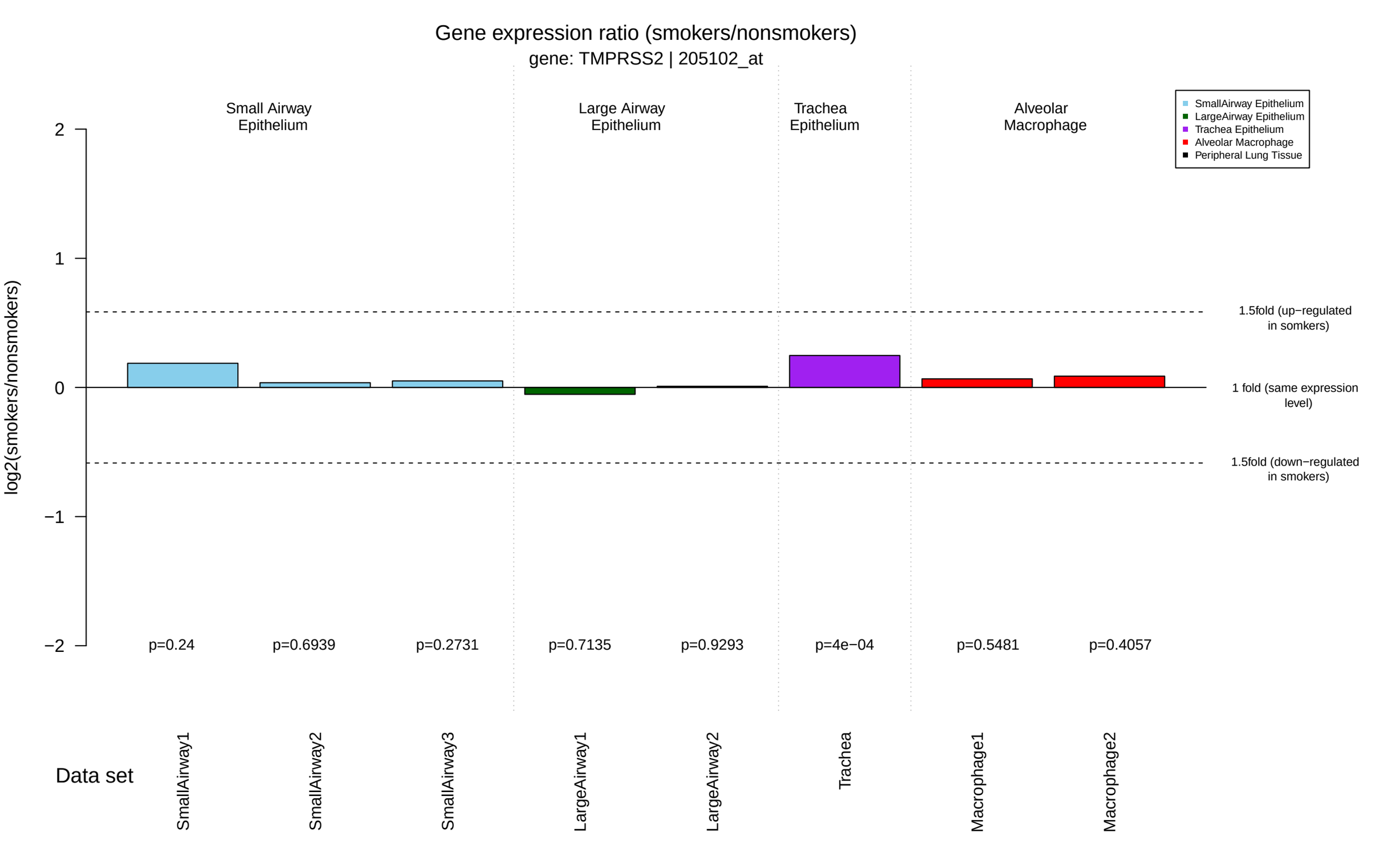
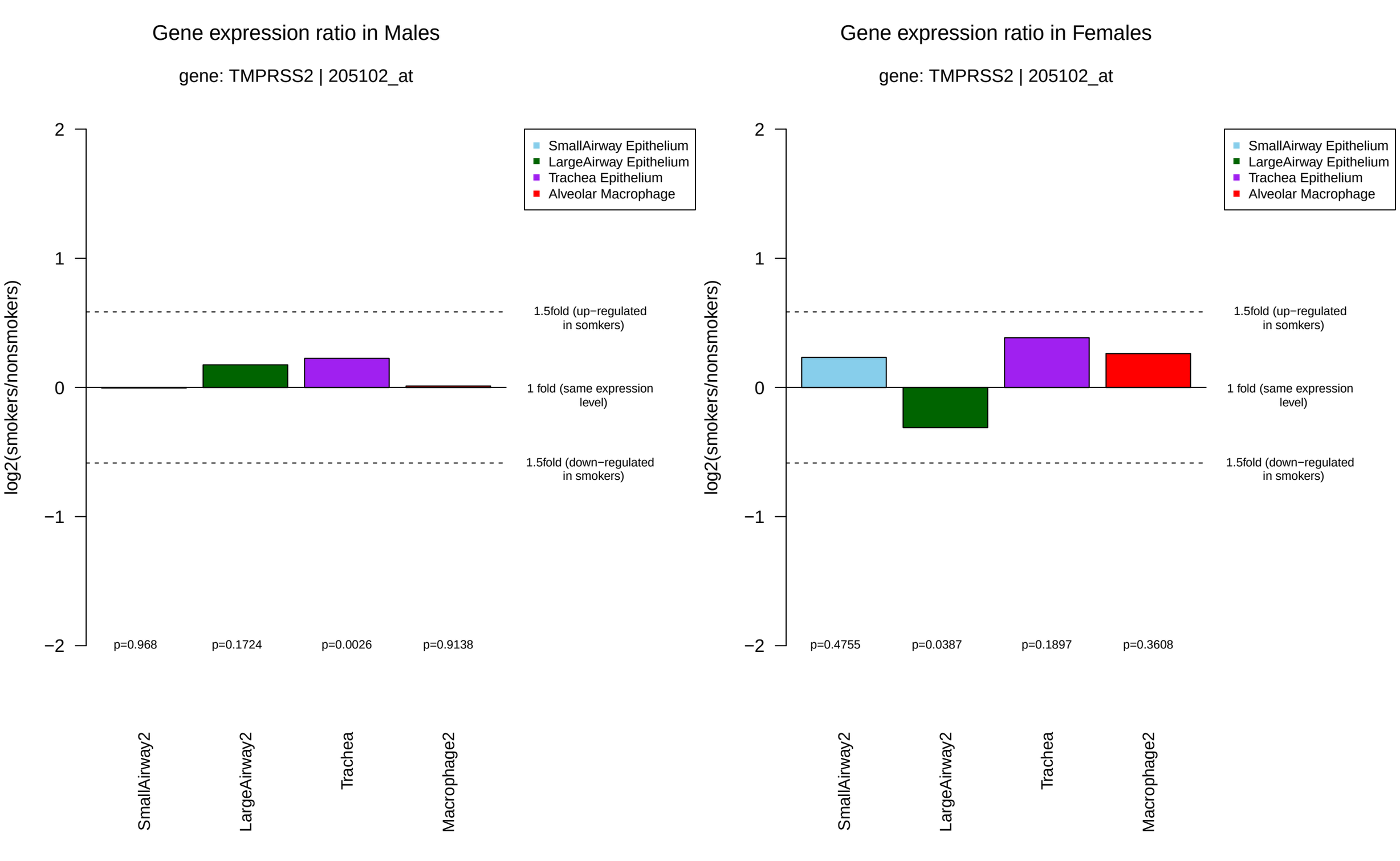
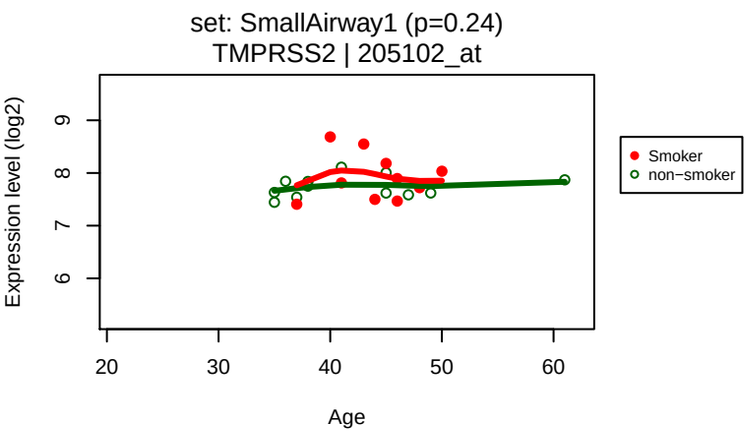
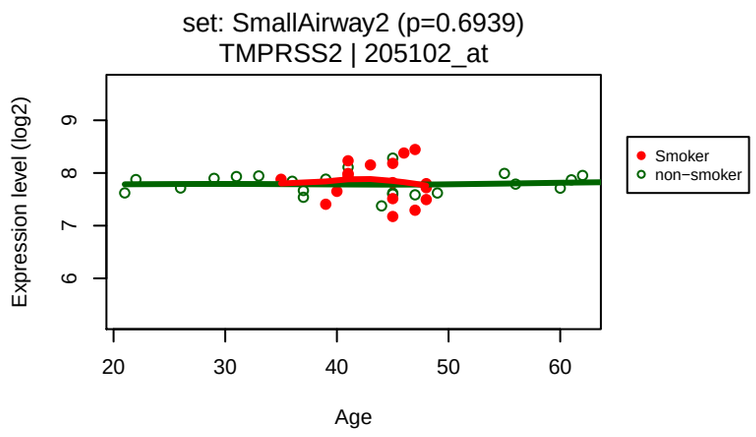
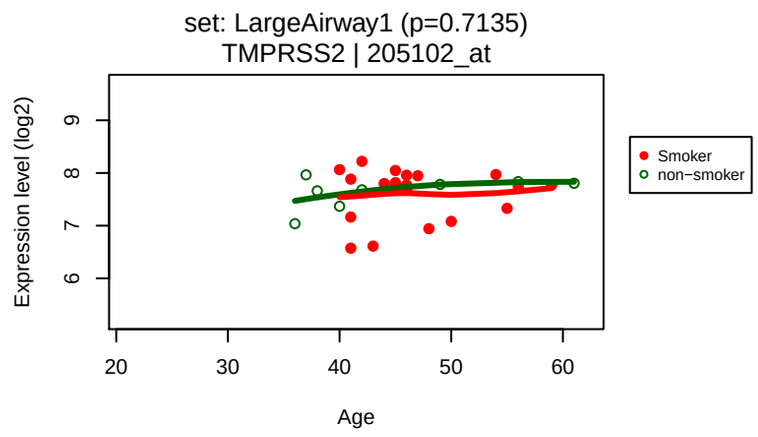

Gene expression of smokers and nonsmokers
gene: TMPRSS2 | 205102_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.24
p=0.6939
p=0.2731
p=0.7135
p=0.9293
p=4e−04
p=0.5481
p=0.4057
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

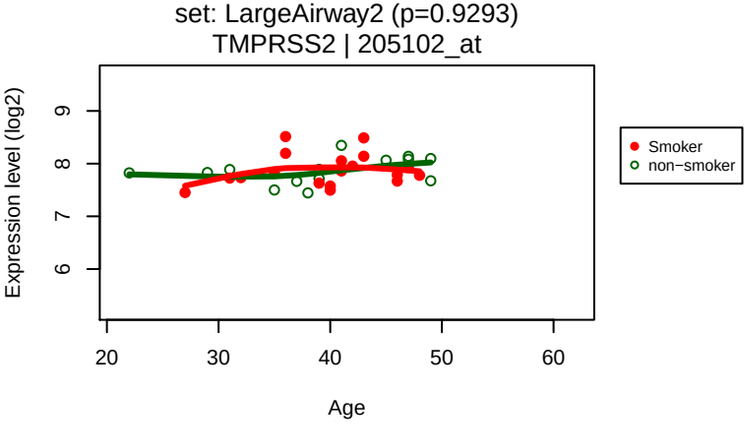

Gene expression ratio (smokers/nonsmokers)
gene: TMPRSS2 | 205102_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.24
p=0.6939
p=0.2731
p=0.7135
p=0.9293
p=4e−04
p=0.5481
p=0.4057
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

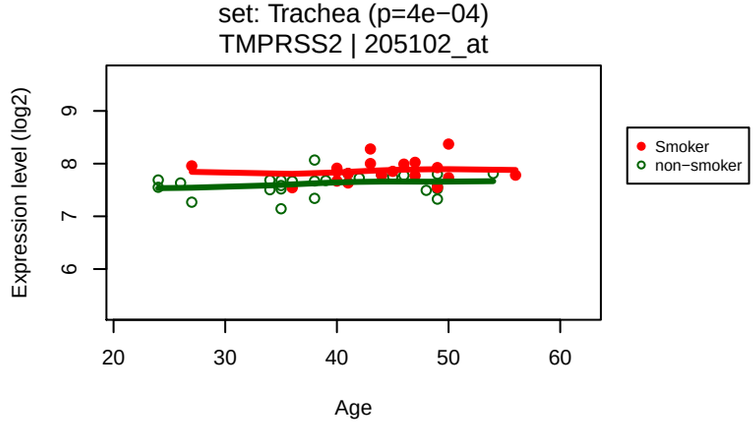

Gene expression ratio in Males
gene: TMPRSS2 | 205102_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.968
p=0.1724
p=0.0026
p=0.9138
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TMPRSS2 | 205102_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4755
p=0.0387
p=0.1897
p=0.3608
SmallAirway2
LargeAirway2
Trachea
Macrophage2

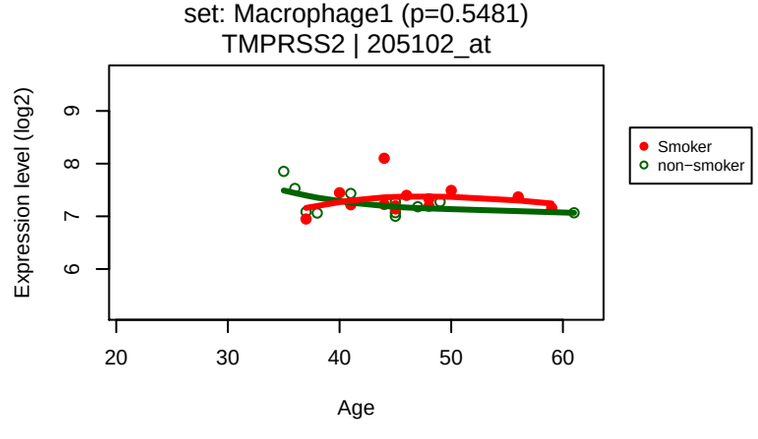

set: SmallAirway1 (p=0.24)
TMPRSS2 | 205102_at
Expression level (log2)
Age
Smoker
non−smoker

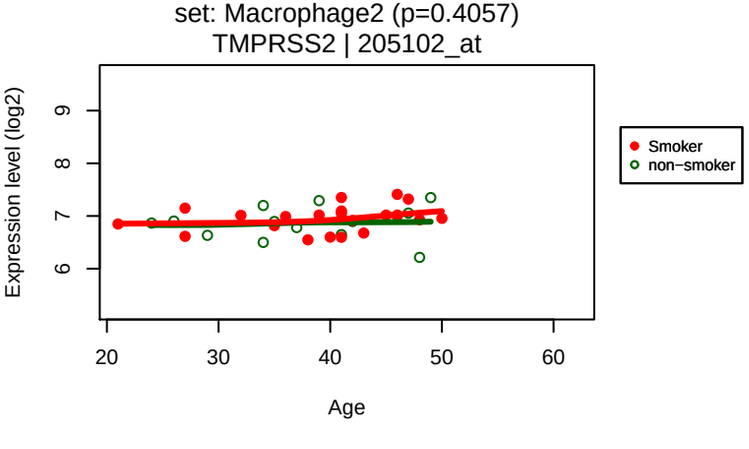

set: SmallAirway2 (p=0.6939)
TMPRSS2 | 205102_at
Expression level (log2)
Age
Smoker
non−smoker

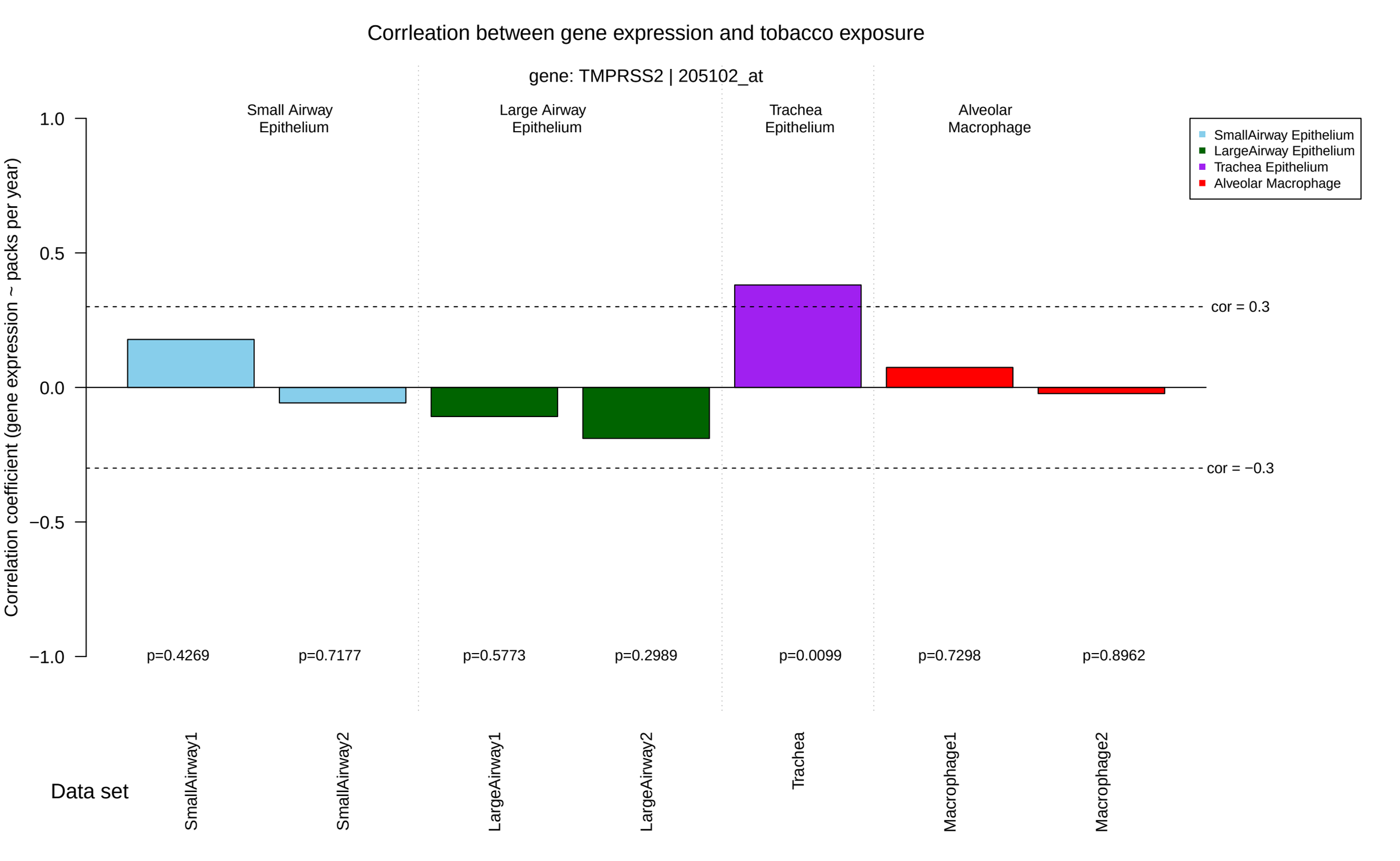

set: LargeAirway1 (p=0.7135)
TMPRSS2 | 205102_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9293)
TMPRSS2 | 205102_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=4e−04)
TMPRSS2 | 205102_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5481)
TMPRSS2 | 205102_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4057)
TMPRSS2 | 205102_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TMPRSS2 | 205102_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4269
p=0.7177
p=0.5773
p=0.2989
p=0.0099
p=0.7298
p=0.8962
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2